



1.1

Change in Tandem

Student Notes and Guided Practice

AP Precalculus - Unit 1

| ECHS Mathematics

| Teacher: Mohammad Abu-

Ghuwaleh

Learning Targets

- Recognize a function across tables, graphs, mappings, formulas, and contexts.
- Determine domain, range, image, and preimage while respecting contextual restrictions.
- Explain how an input quantity and an output quantity change in tandem.

Essential Question

How can we decide whether one changing quantity determines another?



Every input has exactly one output.

Concept Framework

Key Idea

A function assigns exactly one output to each input in its domain.

Key Idea

Domain restrictions may be mathematical, contextual, or both.

Key Idea

A repeated output is allowed; a repeated input with two different outputs is not.

Key Idea

Image means output produced by a stated input; preimage means input producing a stated output.

Formula Map**Formula and Structure**

$f(a)$ = the output when the input is a

Formula and Structure

$\text{Dom}(f) = \{x : f(x) \text{ is defined}\}$

Formula and Structure

$\text{Range}(f) = \{f(x) : x \in \text{Dom}(f)\}$

Worked Examples**Worked Example 1**

Problem. A tank contains $V(t) = 120 - 8t$ liters for $0 \leq t \leq 15$. Find the domain and range and interpret $V(6)$.

Solution. Time is restricted to $[0, 15]$. Since V decreases from 120 to 0, the range is $[0, 120]$. Also $V(6) = 72$: after 6 minutes, 72 liters remain.

Worked Example 2

Problem. Does $x^2 + y^2 = 9$ define y as a function of x on $[-3, 3]$?

Solution. No. For most inputs, $y = \pm\sqrt{9 - x^2}$, so one input produces two outputs. Restricting to $y \geq 0$ would create a function.

Worked Example 3

Problem. For $f(x) = \sqrt{5 - 2x}$, determine the domain and find all preimages of 1.

Solution. Require $5 - 2x \geq 0$, so $x \leq 5/2$. Solve $\sqrt{5 - 2x} = 1$: $x = 2$.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Determine whether $\{(-2, 1), (0, 3), (2, 1), (0, 5)\}$ defines a function. Justify.

2. For $f(x) = \frac{1}{\sqrt{x-2}}$, find the domain.

3. A ride costs $C(n) = 18 + 7n$ QAR for integer $n \in \{0, 1, \dots, 6\}$. State domain and range.

4. For $f(x) = |x - 4|$, find all preimages of 6.

5. Explain why a vertical line graph $x = 3$ is not y as a function of x , but can define x as a function of y .

6. Find the range of $f(x) = x^2 - 6x + 11$ on $[1, 7]$.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.